PROJECT PROPOSAL

A 3D first-person survival horror series for the web

Boon Sop: The Unbroken Debt

បុណ្យសព៖ បំណុលមិនដាច់

Five episodes of Khmer folk horror, set in French-colonial Cambodia in 1932 and playable in any modern desktop browser with no installation.

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Author	Rithy Bondeth
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01 Executive Summary

Boon Sop: The Unbroken Debt (**បុណ្យសព៖ បំណុលមិនដាច់**) is a first-person atmospheric survival horror game that runs directly in desktop web browsers. It is set in Cambodia in 1932, during the French Protectorate, and draws on Khmer folklore, Buddhist ritual and colonial history.

The player is **Sora**, a 24-year-old illustrator from Phnom Penh. A broken phone call from her grandmother brings her home to Battambang, where her uncle has died under strange circumstances. She learns that in 1910 her great-grandfather made a pact with a **Neak Ta** (**អ្នកតា**), a land guardian spirit, and that the debt still binds her family. Across five episodes and five provinces, Sora must find and break five seals of the pact before the debt claims her too.

The game is released as a **five-episode season**. Each episode is 3–4 hours long, for a total of 15–20 hours. It runs on WebGPU with a WebGL2 fallback and is built with **Babylon.js**, so players start by opening a link, with no download, installer or gaming PC required.

Project at a glance

GENRE
First-person survival horror
Atmospheric, narrative, stealth and puzzles

PLATFORM
Desktop web browsers
Chrome, Edge, Firefox, Safari

TECHNOLOGY
Babylon.js
WebGPU with WebGL2 fallback

FORMAT
5 episodes
3–4 hours each · 15–20 hours total

SETTING
Cambodia, 1932
French Protectorate era

AUDIENCE
Psychological horror fans
Players of *Fatal Frame*, *Silent Hill*, folklore horror

Why this game, why now

An unused setting	Horror games regularly draw on Western, Japanese and Korean folklore. Khmer folklore and 1930s Cambodia are almost absent from games, so the setting itself is a hook.
No-friction access	Browser technology now supports high-quality 3D on ordinary laptops. A link is easier to share than a store page and a 50 GB download.
Built for episodes	Each episode is a complete story with its own location, enemy and mechanic, released one at a time. Episode 1 tests the idea with real players before the rest is built.

02 Vision and Design Pillars

The player should feel like a stranger in her own family's history: alone in the dark, carrying a small light, and surviving through knowledge of rituals and taboos rather than through force.

2.1 Design pillars

Pillar	What it means	How it shows up in play
Cultural authenticity	Khmer architecture, ritual, music and belief are the foundation of the horror, not decoration.	Stilt houses, pagodas and funeral customs are researched and reviewed by Cambodian advisors. Rituals solve problems.
Light is life	Sora's kerosene lamp is her only real defence and her biggest risk.	Oil runs out, light attracts some entities, and shutters and matches create constant choices.
Fully diegetic interface	No on-screen HUD. Everything the player knows comes from objects in the world.	Map, notes and inventory are in Sora's satchel and sketchbook; the oil level is visible on the lamp itself.
Knowledge over force	Sora never fights. She survives by understanding the rules of each spirit.	Stealth, sound, line of sight, cultural taboos and counter-rituals replace combat.
Sound tells the truth	In the dark, the ears are more reliable than the eyes.	3D spatial audio lets players locate entities they cannot see.

2.2 Tone

Slow dread over cheap shock. The horror builds through silence, the creak of teak floors, rain on roof tiles and the feeling of being watched. Visually, the game looks like a 1930s photograph come to life: film grain, sepia shadows, high-contrast chiaroscuro lighting and warm flame against cold darkness.

03 Story and Setting

3.1 Series synopsis

In 1910, during famine and a cholera outbreak, Sora's great-grandfather made a pact with a Neak Ta to save his family. The price was the devotion and freedom of his bloodline for four generations. The pact was bound by five sacred objects, the **seals**, hidden across Cambodia.

In 1931, Sora's estranged Uncle Sok tried a forbidden ritual to break the pact and died. In 1932, Sora returns home for his funeral and inherits his unfinished task. Each episode follows a trail of blood across her sketchbook map to the next seal. In the final episode, she faces the Neak Ta itself at a sunken temple on the sacred mountain of Phnom Kulen.

3.2 Protagonist: Sora (សូរ៉ា)

Age and background	24. Left her village in Battambang years ago to work as an illustrator in Phnom Penh.
Personality	Independent, educated and creative. Sceptical of the old beliefs at first, she is forced to learn them to survive.
Core skill	Her artist's eye. She sketches what she sees, so clues, maps and symbols appear in her sketchbook as she discovers them.
Equipment	Leather drafting satchel, sketchbook, fountain pens, brass matchbox, kerosene hurricane lamp. Later a compass and the remnants of the broken seals.
Arc	From an outsider who wants to escape her family's past to the one who carries it, understands it and ends it.

3.3 The antagonist: the Neak Ta

The Neak Ta (អ្នកតា) appears as a towering shadow wearing an ancient brass mask, and it grows stronger as each seal breaks. In Khmer belief, Neak Ta are guardian spirits of the land who are traditionally respected and protective. In this story, it is not evil: it is a guardian enforcing a contract that Sora's family entered and then tried to escape. The true ending reflects this, as the spirit bows and releases the family rather than being destroyed.

04 Episode Roadmap

Each episode moves to a new province and introduces one new core mechanic and one new type of entity, while reusing everything learned before.



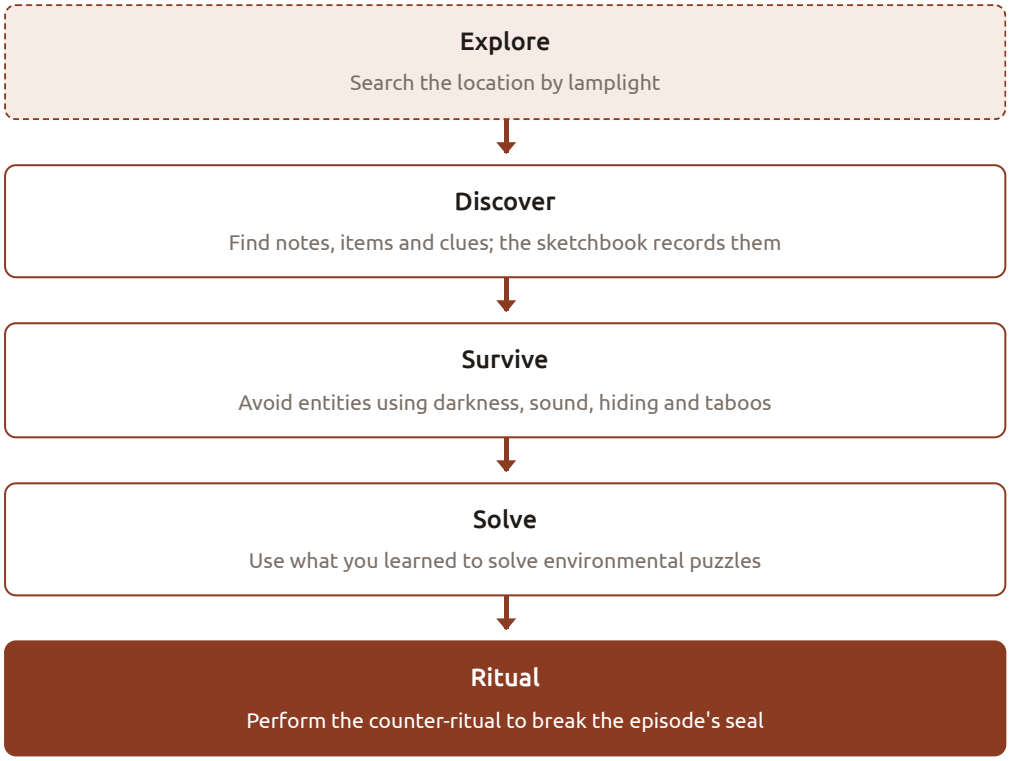
Episode	Setting	Core mechanic	Entity	Seal
1 The House of Mourning ផ្ទះបុណ្យ	Teak stilt house, Battambang	Resource management: kerosene, incense, matches	Neak Ta shadow	Bronze Token
2 The Whispering Pagoda វត្តខ្សឹបខ្សៀវ	Village pagoda and stupa yard	Sound-based stealth and line of sight	Pret (hungry ghosts)	Silver Betel Box
3 The Drowned Village ភូមិលិចទឹក	Flooded clinic, Tonle Sap	Boat navigation, wetness and warmth	Chhlong Tonle (water spirit)	Terracotta Water Jar
4 The Rubber Plantation ចំការកៅស៊ូ	Colonial estate, Kampong Cham	Compass navigation and the "don't look back" taboo	Khmaoch Prey (voice mimics)	Iron Tapping Knife
5 The Sunken Sanctuary ប្រាសាទលិចទឹក	Catacombs, Phnom Kulen	Elemental trials and the severance ritual	Neak Ta Guardian	Golden Inscription Slab

4.1 Episode summaries

Ep 1 · The House of Mourning	Sora keeps the night vigil over Uncle Sok's coffin. She must keep her lamp fuelled, the incense burning and the sacred thread around the funeral tent unbroken, while something walks the floorboards above. She finds the Bronze Token and burns it in a counter-ritual before 3:00 AM.
Ep 2 · The Whispering Pagoda	On Pchum Ben, the Festival of the Dead, blind hungry ghosts hunt by sound among the stupas. Sora makes offerings at her ancestors' stupas to reach the scripture library, where a vision reveals the 1910 pact and she recovers the Silver Betel Box.
Ep 3 · The Drowned Village	In a monsoon storm, Sora rows to an abandoned clinic on stilts haunted by the spirit of a woman who drowned there. She must stay dry, restore power to the drainage pump and recover the Terracotta Water Jar from a flooded safe.
Ep 4 · The Rubber Plantation	In fog-bound rows of rubber trees, spirits call Sora's name in the voices of her family. She must not answer or turn around. In the colonial manor she uncovers the full truth of the contract, then melts the Iron Tapping Knife in the factory furnace.
Ep 5 · The Sunken Sanctuary	Beneath Phnom Kulen, Sora completes four elemental trials that test every skill she has learned, then performs the Ritual of Severance before the Neak Ta Guardian. Her choices across the series decide the ending.

05 Core Gameplay

5.1 Core loop



5.2 Key systems

System	Description
Light and resources	The kerosene lamp burns oil, matches are limited, and later the lamp gains a shutter that hides the light without putting out the flame. Darkness is safer from some entities and more dangerous in every other way.
Diegetic sketchbook	Sora draws maps, copies inscriptions and writes notes automatically. The player opens the satchel to review clues, solve puzzles and track objectives.
Stealth and perception	Entities sense the player by sight, sound or both. Footsteps change with the surface, crouching and holding breath reduce noise, and thrown objects distract.
Cultural taboos	Each episode has rules drawn from Khmer belief, such as never answering a voice calling your name in the forest at night. Breaking them has immediate consequences and affects the ending.
Puzzles	Physical, grounded puzzles: a gramophone speed dial, offerings at stupas, bronze mirrors reflecting moonlight, generator fuses, factory gears.
Counter-rituals	Each episode ends with a ritual performed under pressure, for example lighting incense while the entity stands directly behind the player.
Persistence and endings	Progress, collected lore and broken taboos carry between episodes and decide which of the endings the player receives.

06 Cultural Foundation

The game's identity comes from Khmer culture. Every element below is researched and reviewed with Cambodian advisors before it appears in the game.

Area	Elements used
Spirits and beliefs	Neak Ta (អ្នកតា, land guardian), Pret (ព្រេត, hungry ghosts), Khmaoch Prey (ខ្មោចព្រៃ, forest spirits), a drowned water spirit
Rituals and festivals	Funeral vigil and cremation, Pchum Ben (ភ្ជុំបិណ្ឌ), rice-ball offerings Bay Ben (បាយបិណ្ឌ), protective white cotton thread, incense, chanting
Architecture	Teak stilt houses (ផ្ទះឈើដង្ហើង), pagodas, stupas (ចេតិយ), scripture libraries on stilts, French-colonial manors, Angkorian sandstone temples
Objects and texts	Palm-leaf manuscripts (ស្លឹករឹត), betel boxes, terracotta jars, brass ritual objects
Music	Pinpeat ensemble (ពិណពាទ្យ), especially the roneat (រនាត), used sparingly in an ominous ambient score
History	1930s French Protectorate: colonial rubber plantations, forced labour, famine and epidemic, rural life

07 Technology and Performance

7.1 Technical stack

Layer	Choice
Engine	Babylon.js (TypeScript): WebGPU rendering with automatic WebGL2 fallback
Physics	Havok (bundled with Babylon.js)
3D format	glTF 2.0 (`.glb`) with Draco or meshopt mesh compression
Textures	KTX2 / Basis Universal, which stay compressed in GPU memory
Audio	Web Audio API through Babylon.js Audio Engine v2: 3D spatial sound and dynamic reverb
Saves	Browser IndexedDB, plus portable save codes for moving between devices and episodes
Build and hosting	Vite build, deployed as a static website

> **CHANGE FROM V1.0**
> Version 1.0 left the engine open between Babylon.js and PlayCanvas. Babylon.js has now been chosen. It is code-first and open source, has mature WebGPU support, and suits a programmer-led solo project without depending on a cloud editor.

7.2 Performance targets

Target	Value
Frame rate (dedicated graphics card)	60 fps at 1080p
Frame rate (integrated laptop graphics)	30 fps at 720p
Draw calls per frame	120 or fewer, using instancing and merged static meshes
Visible polygons per frame	150,000 or fewer, using level of detail (LOD)
Initial download per episode	85 MB or less, with the first playable area under 20 MB

7.3 Browser support

Chrome and Edge use WebGPU. Safari uses WebGPU on macOS 26 and later. Firefox uses WebGPU on Windows and Apple Silicon Macs. All other desktop browsers fall back to WebGL2. Mobile devices are not supported in the first release.

08 Production Plan

The project is developed by a **solo developer**. Art and audio come from licensed asset libraries for generic objects and custom work for everything culturally specific. The plan is to build and release **Episode 1 first**, learn from players, and then produce the remaining episodes on top of the same systems.

8.1 Episode 1 roadmap (about 34 weeks)

Phase	Name	Length	Main outcome
0	Pre-production	3 weeks	Story gaps resolved, Episode 1 beat sheet, art bible, tools installed
1	Technical spikes	4 weeks	Risky technology proven: controller, lighting, asset pipeline, audio, saves
2	Greybox	5 weeks	All of Episode 1 playable with placeholder shapes
3	Core systems	7 weeks	Reusable, data-driven gameplay systems
4	Art and audio	10 weeks	Final environments, characters, sound and lighting
5	Polish and release	5 weeks	Playtesting, cultural review, browser testing, public release

Episodes 2–5 are estimated at 4–6 months each after Episode 1, because they reuse its systems. The full season is a multi-year project for one person. Releasing episode by episode keeps it achievable and lets each release fund and inform the next. Details are in the *Development Plan* document.

8.2 Roles

The original proposal listed five roles. As a solo project, they are covered as follows:

Role	How it is covered
Lead web game developer	Solo developer
Technical designer and scripter	Solo developer
3D environment and prop artist	Solo developer (Blender) plus licensed asset libraries; commissioned work for hero assets if the budget allows
Sound designer and composer	Self-recorded foley and libraries; Pinpeat music commissioned or licensed from Cambodian musicians
Writer and cultural consultant	Writing by the developer; review by one or two Cambodian cultural advisors at fixed milestones

09 Risks and Mitigation

Risk	Impact	Mitigation
Scope too large for one person	Critical	Release Episode 1 on its own first; build systems once and reuse them; keep a cut list of features to drop first.
Browser memory and crashes	High	Load and unload assets per area; KTX2 textures; memory profiling during every test cycle.
Slow loading	High	Lazy-load assets per area; stream audio; keep the initial load under 85 MB and the first area under 20 MB.
Performance on laptops	High	Test weekly on an integrated-graphics laptop; one shadow-casting light; baked lighting; quality presets.
Cultural inaccuracy or misrepresentation	High	Cambodian advisors review the story, the Episode 1 greybox and the release candidate; Khmer language checked by native speakers.
Browser differences	Medium	Always maintain the WebGL2 fallback; test Chrome, Edge, Firefox and Safari before every release.
Story inconsistencies across episodes	Medium	Open story questions are tracked in the Development Plan and resolved before each episode is built.

10 Conclusion and Next Steps

Boon Sop: The Unbroken Debt brings Khmer folklore to a global horror audience through the most accessible platform there is: a web link. Its diegetic interface, light-based survival and ritual-driven puzzles give it a distinct identity, and the episodic structure makes it achievable for a solo developer.

Immediate next steps

1. Resolve the story questions marked "before Episode 1" in the Development Plan.
2. Write the Episode 1 beat sheet and a one-page art bible.
3. Find one or two Cambodian cultural advisors.
4. Install the toolchain and run the technical spikes.
5. Build the Episode 1 greybox and playtest it.

A Appendix: Document Set

No.	Document	Purpose
00	Project Proposal	This document: vision, story, scope and plan overview
01	Episode 1 · The House of Mourning	Full design and narrative specification
02	Episode 2 · The Whispering Pagoda	Full design and narrative specification
03	Episode 3 · The Drowned Village	Full design and narrative specification
04	Episode 4 · The Rubber Plantation	Full design and narrative specification
05	Episode 5 · The Sunken Sanctuary	Full design and narrative specification
06	Development Plan	Roadmap, architecture, systems, assets, resources and open questions